Input:

Activities Terminal Mar 24 11:29

user07@jannani:~/pc — /usr/bin/vim p.c

```c
#include <stdio.h>
#include<pthread.h>
#include<semaphore.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in=0 ;
int out=0;
sem_t empty;
sem_t full;
sem_t mutex;
void *producer(void *arg)
{
        int item;
        for(int i=0;i<BUFFER_SIZE; i++)
        {
                item = i;
                sem_wait(&empty);
                sem_wait(&mutex);
                buffer[in]=item;
                printf("Produced :%d\n",item);
                in=(in+1)%BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&full);
        }
}
void *consumer(void *arg)
{
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                sem_wait(&full);
                sem_wait(&mutex);
                item=buffer[out];
                                                                        1,18          Top
```

Activities Terminal Mar 24 11:30

user07@jannani:~/pc — /usr/bin/vim p.c

```c
                sem_post(&full);
        }
}
void *consumer(void *arg)
{
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                sem_wait(&full);
                sem_wait(&mutex);
                item=buffer[out];
                printf("Consumed: %d\n",item);
                out=(out+1)%BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&empty);
        }
}
int main()
{
        pthread_t p,c;
        sem_init(&empty,0,BUFFER_SIZE);
        sem_init(&full,0,0);
        sem_init(&mutex,0,1);
        pthread_create(&p,NULL,producer,NULL);
        pthread_create(&c,NULL,consumer,NULL);
        pthread_join(p,NULL);
        pthread_join(c,NULL);
        sem_destroy(&empty);
        sem_destroy(&full);
        sem_destroy(&mutex);
        return 0;
}
S
-- INSERT --                                                            54,2          Bot
```

Output:

```
Activities   Terminal                    Mar 24  11:31
                          user07@jannani:~/pc
[user07@jannani pc]$ vi p
[user07@jannani pc]$ vi p.c
[user07@jannani pc]$ gcc p.c
p.c: In function ‘consumer’:
p.c:29:23: error: ‘BUffER_SIZE’ undeclared (first use in this function); did you mean ‘BUFFER_SIZE’?
   29 |         for(int i=0;i<BUffER_SIZE;i++){
      |                       ^~~~~~~~~~
      |                       BUFFER_SIZE
p.c:29:23: note: each undeclared identifier is reported only once for each function it appears in
[user07@jannani pc]$ vi p.c
[user07@jannani pc]$ vi p.c
[user07@jannani pc]$ gcc p.c
[user07@jannani pc]$ ./a.out
Produced :0/nProduced :1/nProduced :2/nProduced :3/nProduced :4/nConsumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[user07@jannani pc]$ vi p.c
[user07@jannani pc]$ gcc p.c
[user07@jannani pc]$ ./a.out
Produced :0
Produced :1
Produced :2
Produced :3
Produced :4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[user07@jannani pc]$ vi p.c
[user07@jannani pc]$
```